

# **Title of dissertation or thesis**

by

**Dean Spencer Mills**

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# SUMMARY

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by

**Dean Spencer Mills**

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## LIST OF ABBREVIATIONS

|      |                              |
|------|------------------------------|
| ANN  | artificial neural network    |
| EMI  | electromagnetic interference |
| MCMC | Markov chain Monte Carlo     |
| SVM  | support vector machine       |

**Note:** Make sure the list of abbreviations is sorted alphabetically. This list is formatted as a table with no borders. Insert rows as needed.

## LIST OF SYMBOLS [OPTIONAL]

|     |            |
|-----|------------|
| $I$ | current    |
| $R$ | resistance |
| $V$ | voltage    |

**Note:** Make sure the list of symbols is sorted alphabetically, and/or according to the Greek alphabet. The list is formatted as a table with no borders. Insert rows as needed.

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# CHAPTER 1 INTRODUCTION

This chapter motivates multi-person pose grouping as a separable problem within bottom-up human pose estimation (HPE), identifies the gap in modular detector-agnostic grouping methods, and sets out the research questions, approach, and contributions that the rest of the dissertation defends.

## 1.1 CONTEXT

Introduces the detection/grouping split in bottom-up HPE: modern detectors (HigherHRNet, HRNet) produce accurate per-joint heatmaps; assigning those joints to person identities is the remaining bottleneck. Summarises why joint-to-person assignment is intrinsically hard (unknown  $K$ , type constraints, occlusion, scale) and why it deserves to be studied as a module in its own right.

## 1.2 PROBLEM STATEMENT

Formulates the task: given  $N$  detected keypoints with joint-type labels  $t_i \in \{1, \dots, 17\}$  from an image containing an unknown number of people  $K$ , assign each keypoint to one of  $K$  person groups under the constraint that every person has at most one keypoint of each type.

## 1.3 RESEARCH GAP

Existing grouping methods (associative embeddings in HigherHRNet, part-affinity fields in OpenPose, HGG, CenterGroup) are tightly coupled to their host detectors and trained jointly with them. A *modular* grouping module that operates on any detector’s keypoint output is largely unstudied. This is the gap this work addresses.

## 1.4 RESEARCH OBJECTIVE AND RESEARCH QUESTIONS

The objective is to design and evaluate a detector-independent, GNN-based pose grouping module. The investigation addresses the following research questions. Bracketed short answers anticipate the findings from Chapter 4; full support is provided there.

- **RQ1 — The lever.** Is the bottleneck for clustering-based pose grouping the embedding, or the grouping algorithm? *[Answer: the embedding. Four learned grouping heads (DEC, slot attention, graph partitioning, SA-DMoN) fail to beat plain kNN on the same embeddings; strengthening the embedding (PG-GAT) then beats every prior configuration.]*
- **RQ2 — Inductive biases.** Which skeleton-aware modifications to a GATv2 embedding improve pose grouping, and by how much? *[Answer: three modifications — type-pair attention modulation, skeleton-aware positional encoding, and a type-repulsion loss term — each contribute independently; the full combination is the best configuration.]*
- **RQ3 — Detector independence.** Can a grouping module trained separately from the detector match an integrated end-to-end grouping mechanism (HigherHRNet’s associative embedding head) on a real-image benchmark? *[Answer: not fully. PG-GAT + kNN end-to-end is approximately 0.04 PGA below HigherHRNet AE on COCO, with detector-independence as the trade-off.]*
- **RQ4 — Embedding geometry.** Does the choice of embedding space (Euclidean vs hyperbolic) affect pose grouping quality? *[Answer: yes before COCO fine-tuning, where hyperbolic at  $d = 32$  matches Euclidean at  $d = 128$  (an efficiency edge). After fine-tuning the advantage disappears — reported as an efficiency trade, not a quality win.]*
- **RQ5 — Graph primitive.** Is a higher-order graph primitive (skeleton triplets) a better node for pose grouping than single joints? *[Answer: no. TriGAT underperforms joint-level PG-GAT on every comparison, with the voting step (triplet clusters to joint labels) as the dominant failure mode.]*
- **RQ6 — Fine-tuning.** How much does COCO fine-tuning lift a synthetically-trained embedding, and does it damage performance on the synthetic domain? *[Answer: +7.5 PGA on COCO (kNN), the single largest gain of any experiment in this work; synthetic performance is not damaged ( $0.910 \rightarrow 0.928$ ).]*

## 1.5 APPROACH

Train a graph attention network with contrastive loss on detected keypoints; cluster at test time with kNN. The iteration order mirrors the investigation: standard GAT baseline  $\rightarrow$  a series of learned grouping heads (DEC, slot attention, graph partitioning), all of which failed to beat kNN  $\rightarrow$  SA-DMoN, a skeleton-aware modularity objective, which also failed  $\rightarrow$  PG-GAT, which keeps the clustering head simple but strengthens the embedding with three skeleton-aware modifications. Evaluate on synthetic

data (controlled  $K$ , perfect GT) and COCO val2017 (zero-shot and fine-tuned). Close the loop by integrating with HigherHRNet for end-to-end validation.

## 1.6 RESEARCH CONTRIBUTIONS

The contributions of this work are:

1. **PG-GAT**, a skeleton-aware graph attention network for pose-keypoint embedding. Three modifications over standard GATv2: type-pair attention modulation, skeleton-aware positional encoding, and a type-repulsion term in the contrastive loss.
2. **Empirical evidence that the embedding, not the grouping head, is the decisive lever** for clustering-based pose grouping. Documented across four learned-head approaches (DEC, slot attention, graph partitioning, and ten SA-DMoN variants), none of which outperformed plain kNN on the same embeddings.
3. **A COCO fine-tuning protocol** that lifts PG-GAT to 0.957 PGA on COCO val2017 with kNN clustering, and end-to-end to 0.942 when paired with HigherHRNet detection.
4. **An alternative-embedding-space study** (hyperbolic PG-GAT in the Lorentz model): hyperbolic at  $d = 32$  matches Euclidean at  $d = 128$  pre-fine-tune, but the advantage disappears after COCO fine-tuning — documented as an efficiency trade, not a quality win.
5. **An alternative-primitive study** (TriGAT, triplet nodes): a negative result reported with the failure-mode analysis (voting step propagates errors).

## 1.7 RESEARCH OUTPUTS

The primary research output is a journal paper on PG-GAT, targeted at *Image and Vision Computing* (Q1, IF 5.34), with *Pattern Recognition* (Q1, IF 9.84) as a stretch target.

## 1.8 OVERVIEW OF STUDY

Chapter 2 reviews the literature on bottom-up human pose estimation, graph neural networks, and clustering methods relevant to pose grouping. Chapter 3 presents the mathematical preliminaries (GNN, GCN, GAT/GATv2, DEC, slot attention, graph partitioning, DMoN) and describes the methodology behind each method investigated, culminating in PG-GAT. Chapter 4 reports experimental results across the methods. Chapter 5 discusses what the results tell us about the problem. Chapter 6 concludes and indicates future work.

## 1.9 DECLARATION REGARDING USE OF GENERATIVE AI

Please declare if and how generative AI was used in this work. This must be done following the steps (adapted for dissertations/theses) outlined in 8.2.1.B.10 in the IEEE Publication Services and Products Board Operations Manual<sup>1</sup>:

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<sup>1</sup><https://pspb.ieee.org/images/files/PSPB/opsmanual.pdf>

## **CHAPTER 2   LITERATURE REVIEW**

This chapter situates multi-person pose grouping within the broader human pose estimation literature, reviews the existing families of grouping mechanisms, and surveys the graph neural network and representation-learning work that PG-GAT builds on. The review closes by identifying the specific gap that this dissertation addresses.

### **2.1   MULTI-PERSON HUMAN POSE ESTIMATION**

Overview of top-down (detect-person-then-pose) versus bottom-up (detect-joints-then-group) pipelines. Why bottom-up is attractive (constant time in number of people, robust to overlapping detections) and why it reintroduces the grouping problem. Canonical detectors: Hourglass networks, HRNet, HigherHRNet.

### **2.2   GROUPING IN BOTTOM-UP POSE ESTIMATION**

Review of the major grouping mechanisms in the bottom-up literature: part-affinity fields (OpenPose), associative embeddings (Newell et al.), PersonLab mid-range offsets, HGG (graph-based), CenterGroup. How each binds tightly to its host detector and why that limits modularity.

### **2.3   GRAPH NEURAL NETWORKS FOR POSE**

Existing graph-based approaches in the pose literature: STGAT (skeleton-based action recognition), GAST-Net (single-person 3D pose from video), skeleton-aware GCN variants. How these differ from the multi-person keypoint grouping task — none of the existing “skeleton-aware” networks target grouping, which motivates both the PG-GAT contribution and the naming choice (avoiding collision with STGAT/GAST-Net descriptors).

### **2.4   REPRESENTATION LEARNING FOR CLUSTERING**

Contrastive learning (InfoNCE, SimCLR), triplet losses, metric learning. Deep clustering lineage: DEC (Xie et al.), IDEC, DeepCluster, DMoN. What these methods assume about the feature space and how those assumptions interact with the pose-grouping task.

## **2.5 GEOMETRIC EMBEDDING SPACES**

Euclidean versus non-Euclidean embedding spaces. Hyperbolic embeddings (Poincaré ball, Lorentz model) and their suitability for tree-structured data. Relevance to skeletons, which are trees. Sets up the motivation for the hyperbolic PG-GAT variant evaluated in Chapter 4.

## **2.6 RESEARCH GAP**

Synthesises the review: modular grouping modules for bottom-up HPE are under-explored; existing graph-based pose networks target different tasks; the clustering-head literature has not been systematically applied to pose grouping; and the choice of embedding space (Euclidean vs hyperbolic) is untested in this domain. PG-GAT and its ablations address these gaps.



## CHAPTER 3 METHODOLOGY

This chapter presents the mathematical preliminaries that every method in this dissertation builds on, the problem formulation and data pipeline, and the methodology for each approach investigated — from the standard GAT baseline, through the series of learned grouping heads that were tried and failed, through SA-DMoN, and finally to PG-GAT. Two variants (hyperbolic PG-GAT, TriGAT) and the integration with a realistic detector are described last.

### 3.1 PRELIMINARIES — GRAPH NEURAL NETWORKS

#### 3.1.1 The message-passing framework

General formulation of graph neural networks as iterated message passing. Nodes, edges, neighbourhoods, aggregation, update. Why the message-passing view unifies GCN, GAT, and most variants used in this dissertation.

#### 3.1.2 Graph convolutional networks (GCN)

Derivation from spectral graph theory to the Kipf–Welling simplification. The normalised-adjacency formulation  $\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$ , the layer equation  $H^{(l+1)} = \sigma(\tilde{A}H^{(l)}W^{(l)})$ , and the assumptions it makes about edge uniformity. Why uniform edge weights are a limitation for skeletons where joint-pair relationships differ by type.

#### 3.1.3 Graph attention networks (GAT and GATv2)

Original GAT (Veličković et al., 2018) attention:  $e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [W\mathbf{h}_i \parallel W\mathbf{h}_j])$ , followed by softmax over neighbours. The static-attention limitation (shared neighbour ranking across query nodes). GATv2 (Brody et al., 2022) fix: reorder to  $e_{ij} = \mathbf{a}^\top \text{LeakyReLU}(W[\mathbf{h}_i \parallel \mathbf{h}_j])$ , restoring dynamic attention. All experiments in this work use GATv2 as the attention primitive.

## 3.2 PRELIMINARIES — CLUSTERING AND LEARNED GROUPING

### 3.2.1 $k$ -means and constrained $k$ -means

Classical  $k$ -means objective and Lloyd’s algorithm. Constrained  $k$ -means (must-link / cannot-link) as the family that encodes type constraints. Why  $k$ -means appears throughout this dissertation as the canonical “simple” clustering head, and the basis for PGA scoring with GT  $K$ .

### 3.2.2 Deep embedded clustering (DEC)

Xie et al. (2016) DEC objective: Student’s  $t$ -distribution soft assignments  $q_{ij}$ , target distribution  $p_{ij}$ , KL divergence minimisation. Why DEC was a natural candidate (learn embedding and clustering jointly), and how the alternating refinement works.

### 3.2.3 Slot attention

Locatello et al. (2020) slot attention: iterative attention between a set of learned slots and input features, soft-binding each input to a slot via competitive softmax. Why slot attention suggested itself for pose grouping (slots as person identities) and what the iteration assumes.

### 3.2.4 Graph partitioning (MinCut, NormCut)

Graph-cut objectives: minimum cut, normalised cut (Shi & Malik). Relaxation to continuous assignment matrices and the eigenvalue connection. How graph partitioning was applied as a learned grouping head.

### 3.2.5 Deep modularity networks (DMoN)

Tsitsulin et al. (2023) DMoN objective: Newman modularity as a soft, differentiable community-detection loss. The modularity matrix  $B = A - dd^\top / 2m$  and the collapse-prevention regulariser. Why DMoN was the natural follow-up after the three learned heads failed, and what assumption it makes about the relationship between graph structure and cluster membership (this will matter in Chapter 5).

## 3.3 PROBLEM FORMULATION

Formalises the task. Input: set of  $N$  keypoints with types  $t_i$  and 2D positions  $\mathbf{p}_i$ . Output: a partition into  $K$  groups (one per person) with the type constraint. Evaluation metric: Pose Grouping Accuracy (PGA), defined as the Hungarian-matched joint assignment accuracy between predicted and GT groups.

## 3.4 DATASET PIPELINE

The synthetic data generator (controlled  $K$ , perfect GT, procedural skeletons), the COCO keypoints dataset (val2017 used throughout), and the train/val splits. Depth channel handling (retained or

removed) and why that decision matters for COCO transfer.

### 3.5 BASELINE: STANDARD GAT + CONTRASTIVE + KNN

The baseline architecture: stacked GATv2 layers on a fully-connected graph over detected keypoints, contrastive loss pulling same-person keypoints together and pushing different-person keypoints apart, kNN clustering at test time. The depth-input variant and the no-depth variant. This is the starting point of the narrative.

### 3.6 LEARNED GROUPING HEADS

#### 3.6.1 DEC applied to pose grouping

How DEC was instantiated on top of the GAT embeddings: initialisation strategy, cluster-count handling, training objective combined with contrastive loss.

#### 3.6.2 Slot attention applied to pose grouping

Slot attention head on top of the GAT embeddings, with slots as person identities. Number of slots, initialisation, iteration count, and the binding-to-assignment readout.

#### 3.6.3 Graph partitioning applied to pose grouping

Graph partitioning head on top of the GAT embeddings. Edge-weight construction, normalised-cut objective, and how the soft partition is read out into hard assignments.

### 3.7 SA-DMON — SKELETON-AWARE DEEP MODULARITY NETWORKS

A DMoN-based learned grouping head adapted for skeletons. v1 (learnable  $\sigma$ , clamped  $\sigma$ , median  $\sigma$ , spectral-term variants) and v2 (feature-only adjacency, entropy-only regularisation, full formulation). Ten experiments in total, all tried to fix the modularity misalignment in a different way.

### 3.8 PG-GAT — POSE GROUPING GRAPH ATTENTION NETWORK

#### 3.8.1 Design intuition

Reframes the investigation: every learned head failed to beat kNN on the same embeddings, so the bottleneck is the embedding. PG-GAT strengthens the embedding while keeping the grouping head simple (kNN).

#### 3.8.2 Type-pair attention modulation

Per-type-pair learnable bias in the GATv2 attention score, so that different joint-type relationships (e.g. left-shoulder-to-right-shoulder vs left-shoulder-to-head) can be weighted independently.

### **3.8.3 Skeleton-aware positional encoding**

A positional encoding scheme that encodes relative skeleton position (not absolute image position), invariant to translation and scale.

### **3.8.4 Type-repulsion loss term**

An auxiliary contrastive term that explicitly pushes same-type keypoints of different people apart (the hard case: two left shoulders from different people must not collapse).

### **3.8.5 Training procedure**

Two-stage training: synthetic pretraining (contrastive loss, all three modifications active), then COCO fine-tuning on detected keypoints (20 epochs). Hyperparameters, schedule, optimiser.

## **3.9 HYPERBOLIC PG-GAT**

A variant that embeds the output into a Lorentz-model hyperbolic space. Euclidean internal attention, Lorentz output projection, logmap for clustering. Motivation: skeletons are trees, and trees embed more faithfully in hyperbolic space with fewer dimensions. The variant tests whether this translates into a quality or efficiency advantage for pose grouping.

## **3.10 TRIGAT — TRIPLET-BASED PRIMITIVE**

A variant where the graph nodes are skeleton triplets (A–B–C chains, 19 per person) rather than individual joints. Pivot-angle features are rotation-invariant articulation descriptors. Triplets are embedded, clustered, and joint labels are voted from triplet-cluster assignments. Included to test whether a higher-order primitive improves the embedding.

## **3.11 INTEGRATION WITH A REALISTIC DETECTOR**

How PG-GAT plugs into a HigherHRNet detection pipeline without joint training. Keypoint detection confidence filtering, the K-estimation head (brief treatment — the full version belongs to the PEng follow-up), end-to-end PGA computation.

## **3.12 EVALUATION PROTOCOL AND METRICS**

PGA definition with Hungarian matching. Synthetic evaluation (100-image virtual test set, perfect GT). COCO evaluation zero-shot and after COCO fine-tuning. End-to-end evaluation using HigherHRNet detections. What kNN-PGA measures, what COP-KM-PGA adds, and which numbers are the native MEng headlines (kNN) versus what is mentioned with a pointer to the PEng paper (type-constrained clustering).

## CHAPTER 4 EXPERIMENTAL RESULTS

This chapter reports experimental results for every method presented in Chapter 3, in the order of the investigation. Each section follows the lab-book pattern (setup, table of numbers, interpretation). Failed approaches (learned heads, SA-DMoN, TriGAT, visual-feature augmentation) are reported with their numbers *and* explanations of why they failed. Honest caveats are flagged where the numbers are close calls.

### 4.1 BASELINE: STANDARD GAT

Results for the standard GAT baseline with and without the depth channel. The key observation: removing the MiDaS depth channel *improved* COCO transfer by roughly four points (MiDaS has a synthetic-to-real domain gap). Numbers: 0.994 synth kNN / 0.838 COCO kNN with depth; 0.901 / 0.879 without.

### 4.2 FAILED LEARNED GROUPING HEADS

#### 4.2.1 DEC

DEC plateaued at 0.654 synth and did not transfer meaningfully to COCO. The Student's- $t$  target distribution concentrates cluster mass but does not separate identities in the pose setting.

#### 4.2.2 Slot attention

Slot attention underperformed kNN on the same embeddings (0.858 synth / 0.788 COCO with the depth baseline). The competitive softmax does not encode the type constraint.

#### 4.2.3 Graph partitioning

Graph partitioning was closest to beating kNN on synthetic (0.954) but collapsed on COCO (0.752). Highly sensitive to edge-weight construction, does not transfer.

#### 4.2.4 Cross-cutting finding

Every learned head failed to beat plain kNN on the same embeddings. This was the first evidence that the lever is the embedding, not the grouping head.

### 4.3 SA-DMON — TEN EXPERIMENTS

Full results table for all ten SA-DMoN variants (v1 with learnable / clamped / median  $\sigma$ , v1 with  $\lambda$  sweeps, v1 without spectral term, v2 with feature-adjacency variants, v2 with entropy-type-only regularisation, v2 full). Every variant plateaus in the 0.77–0.81 synthetic range. Root cause analysis is deferred to Chapter 5, but the empirical picture is: no SA-DMoN formulation closes the gap to kNN.

### 4.4 PG-GAT ABLATIONS

Four-row ablation: type-pair-only, pos-enc-only, repulsion-only, full. Each modification contributes independently; the full variant is the sum. Headline synthetic number: 0.910 synth kNN / 0.882 COCO kNN on the synthetic-only training regime. Table with explicit  $\Delta$  against the standard-GAT-no-depth baseline.

### 4.5 COCO FINE-TUNING

PG-GAT fine-tuned on COCO keypoints (20 epochs). COCO kNN PGA rises from 0.882 to 0.957. Synthetic PGA rises from 0.910 to 0.928 (the fine-tune is not destructive to synthetic performance). This is the single largest gain of any experiment in the dissertation.

### 4.6 HYPERBOLIC PG-GAT

Two-part result. Pre-fine-tune: hyperbolic at  $d = 32$  matches or slightly beats Euclidean at  $d = 128$  (0.918 synth / 0.899 COCO kNN, vs Euclidean 0.910 / 0.882) — consistent with trees needing fewer dimensions in hyperbolic space. Post-fine-tune: the advantage disappears. Hyperbolic  $d = 32$  COCO-fine-tuned reaches 0.943 kNN, still below Euclidean  $d = 128$  COCO-fine-tuned (0.957). Reported honestly as an *efficiency trade* (comparable quality at  $4\times$  smaller embedding), not a quality win.

### 4.7 TRIGAT — NEGATIVE RESULT

Triplet-based primitive underperforms joint-level PG-GAT on every comparison: 0.898 synth / 0.855 COCO with kNN, versus 0.910 / 0.882 for joint-level. Worst failure is with strict assignment (THA): 0.804 synth. Root cause: triplet clusters must be voted back to joint labels, and small triplet-level errors propagate to multiple joint votes. Documented as a negative result.

#### 4.8 END-TO-END INTEGRATION WITH HIGHERHRNET

Replace the associative-embedding tag head in HigherHRNet with PG-GAT. End-to-end PGA using HigherHRNet-W32-512 for detection: PG-GAT + kNN = 0.942, PG-GAT + COP-Kmeans = 0.955 (COP-KM detail deferred to the PEng paper), HigherHRNet native AE grouping = 0.985. The gap to 0.985 comes from the tight coupling of the tag head to its backbone features, not from embedding-quality issues.

#### 4.9 MODEL SIZE AND MODULARITY

Parameter counts: PG-GAT 170,560; K-head 18,625; HigherHRNet-W32 full model 28,645,331; HigherHRNet AE tag head alone  $\sim 561$ . Correction to the earlier “ $168\times$  smaller” claim: PG-GAT is *larger* than the pure AE tag head (170K vs 561). The contribution is modularity — PG-GAT plugs into any detector’s keypoint output without joint training; the AE head cannot.

#### 4.10 VISUAL-FEATURE AUGMENTATION (PG-GAT V2) — NEGATIVE RESULT

Added MobileNetV2 features sampled at each keypoint location as additional node input. Result: 0.968 COCO kNN / 0.964 E2E — a 0.8-point E2E improvement at the cost of 2.2M extra parameters. Total pipeline (30.8M) is larger than HigherHRNet (28.6M) while still scoring lower (0.964 vs 0.985). Reported as a negative result; the Paper 3 idea was dropped on this basis.

## CHAPTER 5 DISCUSSION

This chapter interprets the results of Chapter 4, answers the research questions from Chapter 1, and examines the limitations of the work. The central question of the chapter is *why* the investigated methods succeeded or failed — not just what the numbers were.

### 5.1 WHY THE LEARNED GROUPING HEADS FAILED

Discusses the shared failure mode of DEC, slot attention, and graph partitioning: each added a learned assignment mechanism on top of embeddings that were already close to separable under simple kNN. When the embedding is the bottleneck, stacking a learned head reshapes the assignment space but does not improve separation, and can actively hurt transfer (the learned head overfits to synthetic structure).

### 5.2 WHY SA-DMON FAILED

Root-cause analysis of the ten SA-DMoN experiments. The modularity objective is fundamentally misaligned with pose grouping: the kNN graph encodes *spatial* proximity, but the communities we want are defined by *identity*. The spatial null model either cancels with the spatial adjacency or is miscalibrated, and no null-model reformulation (learnable / clamped / median  $\sigma$ , feature-only adjacency, entropy regularisation) repaired the misalignment. This is a structural, not a tuning, problem.

### 5.3 WHY PG-GAT WORKS

Discusses what each of the three modifications buys, drawing on the ablation numbers from Chapter 4. Type-pair attention lets the network weight joint-type relationships independently. The skeleton-aware positional encoding provides translation/scale invariance that the contrastive loss alone does not enforce. The type-repulsion term addresses the hardest pairs (same-type, different-person keypoints that look locally similar). Together they make the embedding separable enough that kNN works.



## 5.4 WHAT THE HYPERBOLIC RESULT TELLS US

Interprets the dim-32 hyperbolic advantage and its disappearance after fine-tuning. Pre-fine-tune, hyperbolic geometry helps because the model is relying on the geometric prior (trees embed naturally in hyperbolic space). After fine-tuning, the Euclidean model learns a sufficient representation from data, and the geometric prior becomes a constraint rather than a help. The result is a quality-matched efficiency trade, not a quality win.

## 5.5 WHY TRIGAT UNDERPERFORMED

The triplet primitive is semantically richer per-token but introduces a voting step that propagates errors from clusters to joint labels. For decisive matching algorithms (THA), voting is especially fragile because a single mis-voted triplet can cascade across three joints.

## 5.6 COMPARISON WITH HIGHERHRNET AE — THE MODULARITY TRADE-OFF

Discusses the 0.955 vs 0.985 gap. HigherHRNet’s AE head is tiny ( $\sim 561$  parameters) but inseparable from its backbone; it gets quality from the backbone features being trained jointly with the tag. PG-GAT is larger (170K parameters) but detector-agnostic — a capability the AE head cannot offer. The correct comparison is not “who wins on COCO” but “what does each buy.” This framing matters for the modularity contribution claim.

## 5.7 ANSWERS TO THE RESEARCH QUESTIONS

Explicit answers to RQ1, RQ2, RQ3 from Chapter 1, grounded in the numbers from Chapter 4.

## 5.8 LIMITATIONS AND THREATS TO VALIDITY

Known limitations: reliance on detector-supplied  $K$  ( $K$ -estimation is outside the MEng core); synthetic-to-real domain gap mitigated but not eliminated; only one backbone evaluated end-to-end; PGA is Hungarian-matched joint accuracy, not mAP, so comparison to COCO-standard HPE numbers is indirect; COCO val2017 is the only real-image benchmark reported.

## CHAPTER 6 CONCLUSION

This chapter summarises the contributions of the dissertation, consolidates the answers to the research questions, and indicates directions for future work. The PEng follow-up is the main pointer forward, but several MEng-local directions are also noted.

### 6.1 SUMMARY OF CONTRIBUTIONS

Restates the five numbered contributions from Section 1.6, each now grounded in the numbers from Chapter 4. PG-GAT is the main technical contribution. The empirical finding that the embedding dominates the grouping head is the main scientific contribution. The COCO fine-tuning protocol, the hyperbolic variant, and the TriGAT negative result are supporting contributions.

### 6.2 ANSWERS TO THE RESEARCH QUESTIONS

Condensed one-paragraph answers to RQ1 (yes — PG-GAT + kNN reaches 0.957 COCO), RQ2 (all three PG-GAT modifications contribute independently; COCO fine-tuning adds the largest single improvement), and RQ3 (PG-GAT is 0.03 PGA below HigherHRNet AE end-to-end, but is detector-agnostic — the trade-off is modularity for a small quality cost).

### 6.3 FUTURE WORK

Three directions. (1) The PEng follow-up re-frames grouping as structured assignment and develops SCOT, SCOT-KI, and THA — closing on HigherHRNet’s 0.985 with algorithmic rather than embedding changes. (2) A fully hyperbolic PG-GAT architecture (hyperbolic attention, not only hyperbolic output) would test the geometric-prior hypothesis more cleanly than the partial variant evaluated here. (3) End-to-end joint training with a detector would give up modularity to regain the coupling advantage AE exploits, and would quantify how much of the 0.03 gap is fundamental versus avoidable.

## REFERENCES

## APPENDIX A DERIVATION OF HIGHER ORDER MODELS

## A.1 EXAMPLE HEADING

### A.1.1 Subheading example

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